

SASKATOON KERNEN FARM, SK, Canada

WMO: 715131

Lat: 52.150N

Lon: 106.550W

Elev: 510

StdP: 95.35

Time Zone: -6.00 (W06)

Period: 96-05

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -33.5	(c) -30.6	(d) -37.4	(e) 0.1	(f) -33.5	(g) -34.0	(h) 0.2	(i) -30.5	(j) 11.1	(k) -13.7	(l) 10.0	(m) -12.5	(n) 2.9	(o) 150	(p) 0.718

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 12.6	(c) 30.6	(d) 17.7	(e) 28.6	(f) 16.9	(g) 26.8	(h) 16.1	(i) 20.5	(j) 27.0	(k) 19.2	(l) 24.9	(m) 18.0	(n) 23.6	(o) 5.0	(p) 170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.4	14.1	23.7	17.0	12.9	22.0	15.7	11.9	21.0	61.5	27.0	56.8	24.8	52.7	23.7	26.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 10.7	(b) 9.5	(c) 8.5	(e) -36.9	(f) 35.3	(g) 1.9	(h) 2.2	(i) -38.3	(j) 36.9	(k) -39.4	(l) 38.2	(m) -40.5	(n) 39.5	(o) -41.9	(p) 41.1		
(4) 10.7	(5) 9.5	(6) 8.5	(7) -36.9	(8) 35.3	(9) 1.9	(10) 2.2	(11) -38.3	(12) 36.9	(13) -39.4	(14) 38.2	(15) -40.5	(16) 39.5	(17) -41.9	(18) 41.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	2.5	-16.7	-11.3	-6.5	4.0	10.1	15.1	18.3	17.6	11.8	3.5	-4.9	-11.7	(6)
(7)		DBStd	13.32	8.42	6.96	8.12	6.17	4.69	3.82	3.37	4.08	4.49	5.26	6.55	8.10	(7)
(8)		HDD10.0	3551	828	597	511	193	57	3	0	1	32	210	446	673	(8)
(9)		HDD18.3	5878	1087	830	769	430	257	111	42	64	200	462	696	932	(9)
(10)		CDD10.0	817	0	0	0	13	61	157	258	237	86	7	0	0	(10)
(11)		CDD18.3	102	0	0	0	0	2	14	41	42	4	0	0	0	(11)
(12)		CDH23.3	1444	0	0	0	13	53	216	514	556	89	3	0	0	(12)
(13)		CDH26.7	434	0	0	0	4	5	54	154	201	17	0	0	0	(13)
(14)	Wind	WSAvg	4.2	4.0	3.9	4.3	4.8	4.9	4.3	3.8	3.7	4.1	4.2	3.9	4.0	(14)
(15)	Precipitation	PrecAvg	364	17	14	17	21	46	66	58	39	33	18	15	19	(15)
(16)		PrecMax	486	43	32	48	62	150	162	147	91	70	76	40	34	(16)
(17)		PrecMin	236	5	1	4	2	8	12	14	3	5	1	1	6	(17)
(18)		PrecStd	58	9	9	11	15	32	36	33	24	18	17	10	7	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.3	4.2	11.4	24.7	27.3	31.8	32.9	34.5	28.9	22.9	12.0	6.0	(19)
(20)			MCWB	1.4	2.1	6.1	10.9	13.7	18.2	20.0	16.8	14.8	11.3	7.1	3.1	(20)
(21)		2%	DB	0.9	1.9	7.8	20.1	25.0	27.7	30.1	31.1	25.9	18.3	8.0	3.5	(21)
(22)			MCWB	-0.3	0.2	3.8	9.1	12.3	15.2	18.6	16.6	13.8	9.5	4.7	1.4	(22)
(23)		5%	DB	-1.8	0.4	5.6	17.3	22.5	25.8	28.2	28.9	23.2	15.0	5.4	1.3	(23)
(24)			MCWB	-2.9	-0.8	2.7	8.7	11.4	15.0	17.7	17.2	13.0	8.8	3.0	0.0	(24)
(25)		10%	DB	-4.9	-1.5	3.4	14.2	19.9	23.5	26.2	26.6	20.4	12.0	3.5	-0.8	(25)
(26)			MCWB	-5.7	-2.6	1.3	7.4	10.4	14.4	17.4	16.6	12.0	6.9	1.6	-1.8	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.7	2.1	6.6	12.5	16.0	20.2	23.1	21.5	17.0	12.2	7.7	3.3	(27)
(28)			MCDWB	2.8	3.8	10.4	21.2	23.5	26.8	29.8	27.3	24.8	20.8	11.6	5.4	(28)
(29)		2%	WB	-0.4	0.5	4.5	10.8	13.5	18.0	21.2	20.0	15.2	10.6	5.1	1.5	(29)
(30)			MCDWB	0.9	1.7	7.2	18.1	21.3	24.3	28.2	26.7	22.2	16.7	7.6	3.3	(30)
(31)		5%	WB	-2.8	-0.8	2.8	9.4	12.2	16.7	19.8	18.7	14.0	9.0	3.3	0.0	(31)
(32)			MCDWB	-1.9	0.5	5.2	15.5	20.5	22.8	25.6	24.7	20.8	14.2	5.1	1.4	(32)
(33)		10%	WB	-5.7	-2.5	1.3	7.9	11.0	15.4	18.6	17.5	12.9	7.3	1.7	-1.8	(33)
(34)			MCDWB	-4.9	-1.5	3.3	13.9	18.5	21.0	23.9	23.5	18.8	11.5	3.3	-0.7	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	9.9	10.2	10.2	11.7	13.5	11.7	12.6	13.5	12.8	11.5	9.3	9.5	(35)
(36)			MCDBR	11.6	10.4	11.1	16.8	17.7	15.7	15.8	17.5	18.4	17.8	11.7	11.3	(36)
(37)			MCWBR	10.7	9.1	7.8	9.0	8.6	6.8	7.2	7.1	9.0	10.8	8.8	10.0	(37)
(38)		5% WB	MCDWR	11.7	10.3	10.2	14.9	14.8	12.6	14.1	14.3	14.7	16.5	11.1	11.1	(38)
(39)			MCWBR	10.8	9.0	7.6	8.5	7.7	6.7	7.5	7.5	8.1	10.6	8.6	9.8	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.251	0.266	0.297	0.340	0.354	0.374	0.374	0.376	0.322	0.282	0.253	0.219	(40)	
(41)		taud	2.284	2.269	2.336	2.348	2.355	2.349	2.348	2.338	2.474	2.557	2.472	2.372	(41)	
(42)		Ebn at Noon	767	860	901	897	897	879	871	849	866	848	780	769	(42)	
(43)		Edh at Noon	62	85	98	111	116	118	116	111	86	64	51	47	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.11	2.15	3.63	4.79	5.89	5.96	6.28	5.09	3.70	2.17	1.09	0.80	(44)	
(45)		RadStd	0.10	0.13	0.27	0.47	0.46	0.47	0.35	0.38	0.35	0.32	0.13	0.09	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(46)
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (15 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page